

Supplementary materials

What makes an AI-generated scientific answer worth acting on?

Draft · 17 September 2026 · Local evidence supplement

S1. Starting materials and biological questions

All conditions received the same scientific content for each case. Articles were converted to numbered plain text; CSV files retained tabular measurements and metadata. Published analysis scripts and meta-analysis tables were evidence to audit, not authoritative answers. No new biological measurements were collected. Supplementary Table S1, the complete filename inventory with format, size, and SHA-256 hash, accompanies this document as [PACKET-INVENTORY.csv](#).

BET/MYC. Five CSV files provide treatment-by-time expression values; individual-mouse bioluminescence over time; replication survival/event records; a published original-survival reconstruction; and meta-analysis summaries. Three R scripts document expression, survival, and imaging analyses. Replication text, original-article text, and a correction provide experimental context. The expression experiment has five biological repeats per treatment, with technical qPCR triplicates already summarized. The eligible replication survival cohort has 12 mice per arm after the specified exclusions. Later imaging has treatment-dependent missingness; an absent value is not zero disease burden.

The task asked whether BET inhibition should advance on a MYC-mediated rationale. Required analyses included expression contrasts at 1 and 8 hours, survival and restricted mean survival through day 36, and common-time plus longitudinal burden analyses. Restricted mean survival summarizes average survival time accumulated up to a specified horizon. The biological issue is whether MYC suppression mediates the in-vivo benefit, considering existing BRD4 perturbation/chromatin and partial MYC-rescue evidence.

CD47. Five CSV files provide mouse-level tumor weights, serial tumor volumes, hematology, ordinal immune-pathology scores, and meta-analysis summaries. Two R scripts and a readme accompany the replication text. No original full-text article is included. Tumor weights are available for seven IgG controls and six treated animals; hematology includes seven animals in each treatment group plus a separate untreated baseline cohort. One treated animal lacks an endpoint weight. Three control tumors define a post hoc spontaneous-regression sensitivity set, not a replacement randomized comparison.

The task asked whether to advance anti-CD47 monotherapy in this model. Required analyses included tumor-weight differences and uncertainty, baseline volume, regression sensitivity, a hematology contrast, missingness, and the original-versus-replication comparison. Anemia establishes systemic biological activity and a safety concern; it does not establish tumor occupancy or acceptable tolerability. Pathology scores do not directly measure phagocytosis.

IDH/2HG. Four CSV files contain lysate absorbance time courses, engineered-cell metabolite signals, patient metabolite signals, and meta-analysis summaries. Replication and original-article text provide context. The kinetics table has repeat, time, and construct columns; the metabolite tables have sample/type labels and total-ion and extracted-ion signals. These are processed signal tables, not raw instrument spectra. Quality-control and blank rows must be distinguished from biological samples. There are seven biological repeats in the enzyme assay, ten samples per construct/timepoint in the cell experiment, and four patient samples per genotype group.

The task asked how strongly IDH2 R172K supports a neomorphic activity—an enzyme function acquired by the mutant—and what would establish disease-maintenance dependence. It required rate reconstruction with correct time units, cell contrasts at 24 and 48 hours, sensitivity to ion-summary choice, and patient contrasts. Neither an absorbance proxy nor a relative metabolite ratio directly measures metabolic flux; cross-sectional patient association does not establish a therapeutic dependency.

S2. Supplementary Table S2: answers by condition

I summarize the submitted answers below. These are records of what each workflow proposed, not endorsements of every biological claim or experimental design. R1/R2 indicate the two repeats. Scores are original two-grader means, rounded to one decimal. Source memos and grades remain authoritative. [RUN-SCORES.csv](#) preserves the exact scores, individual grader totals, dimensions, flags, and missing artifacts. [ANSWER-SOURCES.csv](#) links each output to its memo or available numerical decision and original grades.

BET/MYC

Question: Does the evidence justify advancing BET inhibition on the proposed MYC mechanism, and which single study resolves the most consequential remaining uncertainty?

Condition; scores R1/R2	Submitted answer and next study
C0; 66.0 / 65.0	Both gave conditional support for further BET investigation while withholding confirmation of MYC-mediated benefit. R1 proposed BRD4 or MYC rescue within a xenograft study; R2 specified inducible MYC add-back, controls for baseline transgene effects, and burden/survival readouts.
C1; 51.0 / 67.9	Both favored further BET investigation while deferring the MYC-specific claim. R1 proposed MYC knockdown with drug treatment; R2 proposed inducible MYC rescue and target-engagement checks. Both mischaracterized the qPCR replicate evidence and omitted existing rescue evidence.
C2; 87.1 / 84.5	Both withheld advancement on the specific MYC mechanism. R1 proposed MYC rescue/epistasis with burden and survival; R2 prioritized tumor MYC pharmacodynamics with active/inactive compound controls and an optional perturbation arm. Both supplied strong quantitative reconstructions but omitted existing mechanistic rescue evidence.
CS; 69.9 / 77.1	Both gave qualified support for further BET development with MYC mediation unresolved. Both proposed MYC-rescue xenograft studies connecting expression, burden, and survival. Important numerical or inferential defects remained despite this broadly appropriate direction.

Interpretive point. The workflows often proposed similar study families despite a 26.4-point span in condition means. Correctly naming “rescue” is insufficient: an answer must recognize rescue evidence already available, specify the unresolved in-vivo extension, and avoid declaring mechanism absent solely because a contrast is nonsignificant. The original graders deducted points for C2’s omitted mechanism evidence without issuing critical flags; the flag screen therefore missed a consequential coverage weakness.

CD47

Question: Is anti-CD47 monotherapy supported in the supplied syngeneic breast-tumor model, or which resolving study should precede advancement?

Condition; scores R1/R2	Submitted answer and next study
C0; 81.5 / 90.3	Both declined advancement. R1 proposed a larger longitudinal study with macrophage depletion; R2 proposed prospective regression tracking, untreated controls, and a phagocytosis assay. Both overstated what anemia established about engagement or tolerability.
C1; 81.6 / 87.3	Both deferred advancement. R1 proposed a larger controlled replication with tumor growth and phagocytosis; R2 proposed dose/variant comparisons with macrophage depletion to separate efficacy from hematologic effects. Critical engagement/tolerability claims persisted.
C2; 92.1 / 89.1	Both declined advancement pending a resolving study. R1 proposed a randomized replication with growth, blood, and immune readouts. R2 added missing-outcome handling and antibody-clone comparisons, but described an unsupported original two-clone cocktail. Both received critical flags.
CS; 87.6 / 88.9	Both declined advancement. R1 proposed additional untreated controls, serial growth measurements, and exposure/occupancy checks; R2 proposed dose-response and tumor phagocytosis measurements with longitudinal hematology. R1 had no critical flag; its treatment of anemia still deserves scrutiny.

Interpretive point. All eight recommendations avoided immediate advancement, but that agreement did not prevent errors that could compromise the resolving study. Direct tumor exposure/occupancy, immune effector activity, tumor response, and blood safety answer different questions. Listing one as a proxy for all four leaves failure difficult to interpret.

IDH/2HG

Question: How strong is evidence for IDH2 R172K neomorphic activity, and which experiment is needed before claiming an established disease-maintenance dependency?

Condition; scores R1/R2	Submitted answer and next study
C0; 88.3 / 85.4	Both supported altered biochemistry but withheld disease-maintenance dependence. R1 proposed selective inhibition plus orthogonal perturbation, wild-type controls, and 2HG add-back; R2 proposed inhibition or reversion with functional disease readouts and catalytic controls.
C1; 89.6 / 87.5	Both supported neomorphic activity and treated patient findings as association. Both proposed perturbing mutant activity in established disease and checking whether 2HG reduction accompanied reversal of a disease phenotype. C1 had the highest mean total for this case.
C2; 94.3 / 77.1	R1 supported the biochemical claim and proposed endogenous-dosage inhibition or genetic reversion with wild-type controls and fitness/differentiation readouts. R2 delivered no decision memo. Its JSON nevertheless distinguished biochemistry from dependence and proposed selective perturbation with active/inactive rescue controls; this did not substitute for the requested integrated memo.
CS; 93.1 / 80.6	Both supported neomorphic activity while withholding a maintenance claim. R1 proposed inhibition with 2HG supplementation; R2 specified perturbation after disease establishment, rescue, and explicit inconclusive-outcome handling. R2 lacked the required analysis script.

Interpretive point. IDH-C0-R01 earned full next-study points from both graders. This supports a bounded statement about its rubric-rated design, not wholesale endorsement. For example, rescue delivery and target inhibition must themselves be validated before failed rescue or absent phenotype change can resolve mechanism. The broad distinction between biochemical activity and disease dependence was explicitly requested in the task; agreement does not demonstrate that models would discover that distinction without prompting.

S2a. What higher quality looked like

A calculation that answers the required biological contrast. In [BET-C2-R01](#), the active/inactive MYC expression ratio was 0.154 at 1 hour, with a 95% interval of 0.118–0.201 and five biological repeats per arm. The analysis estimated how much expression differed between active and inactive compound, with uncertainty on that same quantity. [BET-C0-R01](#) emphasized expression relative to the active compound's own pretreatment baseline and standardized effects. Its submitted results did not provide an equivalent active/inactive fold-ratio interval. A baseline decrease and an active/inactive difference are distinct questions, even when both support suppression. What improved was the alignment of the estimate, comparator, and uncertainty with the scientific question.

Both graders awarded C2 full Q1/Q2 credit and deducted C0 points. Their item-level explanations are available for [C2, A, C2, B, C0, A](#), and [C0, B](#). C2 also compared a complete-observation day-7 burden analysis with later survivor-conditioned measurements. This helped expose discordance between molecular and tumor-burden evidence; it did not prove MYC mediation or resolve informative dropout in every model.

A study that can separate explanations. [IDH-C0-R01](#) specified drug and genetic perturbation, a wild-type toxicity control, 2HG add-back, and functional readouts. Concordant effects from different perturbations, followed by rescue of the phenotype when 2HG is restored, would strengthen attribution to loss of 2HG rather than an unrelated drug effect. [IDH-C2-R01](#) already included important endogenous-dosage, wild-type, and functional controls, but offered inhibitor or reversion and omitted explicit rescue/paired orthogonal perturbation. A cleaner numerical analysis did not automatically produce the more discriminating study.

C0 received 10/10 next-study points from [both graders](#). C2 received 10/10 from [grader A](#) and 7.5/10 from [grader B](#), who specifically identified the missing rescue and discordant-outcome rules. I can point to the added control and explain what it tests; the graders did not unanimously prefer it. Rescue must restore the relevant exposure, compartment, and timing; failed rescue alone cannot exclude dependency. Neither proposal was experimentally tested.

Scope of the examples. I use these matched first repeats to show what the ratings rewarded. They do not establish that every C2 calculation or C0 study design was better. Condition-wide means include both repeats and all three cases. CS's higher mean evidence score likewise describes the ratings; an unflagged CS report is not guaranteed free of engagement or tolerability overclaims.

S3. Scoring, reproducibility, and interpretation

The [registered protocol](#), task prompts for [BET](#), [CD47](#), and [IDH](#), [rubric](#), and [grader instructions](#) preceded main evaluation. All 48 main grade files contributed to the scores. Model labels were withheld, but writing style and file structure could reveal condition. Candidate allowances were \$1.60 aggregate list-equivalent for C0/C1/C2 and \$3.00 product-reported for CS, with a 15-minute target. These are different accounting bases, not matched actual compute. The registered candidate setting was high reasoning effort, a provider setting for inference-time reasoning. Sonnet grading used medium effort and GPT grading low effort. The graders used the same model families as candidates, so agreement could reflect shared preferences or errors. Fresh grading contexts do not establish statistical independence.

The registered acceptance gate required ≥ 80 total, ≥ 25 quantitative points, successful code replay, and no unresolved critical scientific error. Acceptance was C0: 0/6; C1: 2/6; C2: 3/6; CS: 2/6. This composite mixes scientific and delivery criteria. It should not be interpreted as a direct estimate of useful or correct answers.

A subsequent [completion diagnostic](#) executed all six C0 scripts in clean sandboxes and reproduced 1,683 numeric leaves of their native outputs. Four wrote a different filename from the registered checker's requirement; two included nonnumeric metrics that aborted its normalizer. Separate summary discrepancies persisted, including BET-C0-R01 interval discrepancies and an IDH-C0-R02 metric combining different estimands. Native repeatability does not establish numerical validity or faithful reporting, and original outcomes were not revised.

A post hoc sensitivity subtracts Q5, the eight-point reproducibility component, from every grader's score before averaging. Remaining means are C0 75.9, C1 70.5, C2 79.4, and CS 77.2, each out of 92. This does not repair other grading effects or provide a new primary endpoint. Original and derived tables remain separate.

S4. Questions for the next experiment

Question 1: Does protecting source review improve the final biological decision? Compare the current C2 arrangement with one that reserves completion of a concise source audit and biological challenge before synthesis. Keep total allowance, evidence access, output contract, and quantitative model fixed. Measure whether the intervention reduces consequential omissions and unsupported mechanism claims. This tests a process change, not merely whether a longer answer earns more points.

Question 2: Is the improvement attributable to the quantitative model or its combination with other roles? Retain C0/C1/C2 comparators, and prospectively specify a quantitative-model × workflow comparison if resources permit. A single GPT-based analyst would help distinguish a strong quantitative model from a benefit of mixed-model teamwork. A factorial comparison varies these factors separately. Repeated runs should remain nested within independently selected paper families.

Question 3: Would experts act on the recommendation? Have blinded biologists and quantitative-methods reviewers assess whether the proposed study is feasible, distinguishes the leading explanations, and gives interpretation rules for beneficial, null, discordant, and technically failed outcomes. Define material error and disagreement handling before reviewing results. Select the case count using pilot variability and a meaningful improvement threshold, rather than treating more repeats on the same three cases as broader validation.

New evaluation families should be withheld from development and include different assays and development decisions. Public-paper recognition and prior training exposure remain possible; recent or appropriately authorized unpublished packets could test transfer without assuming that access restrictions erase prior knowledge. Standardize output validation so interface incompatibility can be reported separately from scientific correctness. Reserve actual prospective laboratory testing for recommendations that pass human review; long-term benefit requires measuring information gained and decisions improved.

These are conceptual extensions, not a frozen follow-up protocol or authorization to run. The previously prepared [prospective follow-up](#) remains [parked](#). Any revised design should be registered before generating its evaluation outputs.

S5. Evidence availability and drafting record

Primary biological references appear in the manuscript. Local scientific anchors are [BET](#), [CD47](#), and [IDH](#); consequential source-grounding examples are documented in the [original source audit](#). These are analysis records, not additional biological studies.

The accompanying files provide exact per-output scores, packet hashes, links to the 24 output records and 48 grades, manuscript section word counts, and verification receipts. The original [aggregate](#) and [quality-analysis code](#) establish the numerical provenance. Supplementary figures depict biological framing and recorded critical flags; main figures depict task-level and component scores. All figures describe existing v4 evidence. Neither an unflagged cell nor a high score certifies scientific correctness.

I used AI assistance to execute and evaluate the study and to prepare this account from its stored evidence. Independent human expert grading and prospective biological validation remain outstanding. This draft adds no candidate runs or biological measurements. Public data/code access remains to be arranged before publication.

Supplementary figures

FIGURE S1 | BIOLOGICAL FRAMING

Three questions about what the evidence permits

Each task asks whether a biological observation justifies a stronger claim and a useful next experiment.

CASE	WHAT THE PACKET SUPPORTS	WHAT REMAINS UNPROVEN	DECISION-CHANGING TEST
BET / MYC	MYC suppression in cells; longer mouse survival. Existing in-vitro rescue evidence also matters.	Does MYC suppression mediate the in-vivo survival benefit?	Extend mechanistic rescue into established disease; control exposure and measure burden + survival.
CD47	Strong red-cell effect; no resolved tumor benefit. Tumor-weight difference: +88 mg [−22, +198].	Did the antibody engage tumor CD47? Could efficacy be separated from toxicity?	Measure tumor exposure / occupancy and effector activity alongside tumor growth and blood counts.
IDH / 2HG	Mutant-linked biochemistry and 2HG accumulation; patient association. Cellular contrast: 527x.	Does established cancer depend on the mutant activity or its product?	Perturb the mutant in established disease; measure fitness and test rescue / orthogonal controls.

Biological framing from the frozen reference, not new experiments. CD47: anti-CD47 – IgG, mg, Welch 95% CI, n=6/7. IDH: 48 h mutant/WT geometric 2HG/glutamate ratio in engineered cells; 95% CI 293–948, n=10/group.

Figure S1. Biological questions underlying the three tasks, based on frozen reference analyses. CD47: treatment-minus-IgG tumor-weight difference with Welch 95% confidence interval, n=6/7. IDH: mutant/wild-type geometric 2HG/glutamate ratio at 48 hours in engineered cells, n=10/group. Proposed tests describe inference goals, not validated experimental protocols.

FIGURE S2 | RELIABILITY AND DELIVERY

High scores still coexisted with consequential defects

A critical flag identifies a claim or design error that could change the recommendation. Unflagged does not mean error-free.

	BET / MYC		CD47		IDH / 2HG		Flagged outputs
	Repeat 1	Repeat 2	Repeat 1	Repeat 2	Repeat 1	Repeat 2	
C0	× Flagged	× Flagged	× Flagged	× Flagged	○ No flag	○ No flag	4/6
C1	× Flagged	× Flagged	× Flagged	× Flagged	○ No flag	○ No flag	4/6
C2	○ No flag	○ No flag	× Flagged	× Flagged	○ No flag	No memo*	2/6
CS	× Flagged	× Flagged	○ No flag	× Flagged	○ No flag	No code*	3/6

*No recorded critical flag, but a required artifact was absent. IDH-C2-R02 lacked decision.md; IDH-CS-R02 lacked analysis.py. Each cell is one output. Counts are affected outputs, not duplicate grader annotations. Original flags and outcomes are unchanged.

Figure S2. An output is flagged if either original grader recorded a critical error. Hatching identifies a missing required artifact. Counts denote affected outputs, not duplicate annotations. Unflagged outputs can retain consequential omissions or overclaims; no new grading was performed for this figure.